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Subject : **Fundamentals of Telecommunication Systems**

Experiment Number : **03**

Experiment Title : Simulation of Signal Transmission through Wired Media

Date of Performance :

Date of Submission :

Experiment No. 03

Title: Simulation of Signal Transmission through Wired Media

Aim

To study and simulate the characteristics of signal transmission through a lossy wired medium (Coaxial Cable) using LTspice and observe the effects of attenuation and propagation delay.

Software Used

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory

Wired Media Characteristics: In telecommunication, physical cables like Coaxial or Twisted Pair are not ideal conductors. They possess electrical properties distributed along their length: Resistance (R), Inductance (L), Capacitance (C), and Conductance (G).

The LTRA Model: The Lossy TRAnsmission line (LTRA) model in LTspice simulates real-world behavior.

Attenuation: The signal strength decreases over distance due to Resistance (R).

Distortion: The signal shape changes (rounding of edges) due to Capacitance (C) and Inductance (L).

Propagation Delay: It takes a finite amount of time for the signal to travel from the source to the destination.

Simulation Procedure

Step 1: Component Selection and Placement

Open LTspice and create a New Schematic.

- *Voltage Source:* Select voltage, and place it. Right-click it, select Advanced, then PULSE. Set: Vinitial=0, Von=5, Tdelay=0, Trise=1n, Tfall=1n, Ton=0.5u, Tperiod=1u.
- *Transmission Line:* Press 'P' search for the component 'ltline'. Select it and place it on the schematic. It will appear with the designator O1 and LTRA.
- *Load Resistor:* Press 'R', place a resistor at the output. Right click on the resistor symbol and set resistance value to 50 ohms (to match the characteristic impedance of co-axial cable).

Step 2: Defining the Cable Model

- Right-click the ltline (O1) component. In the Value field, replace LTRA by typing 'MyCo-axCable'.
- Click the '.t' button (SPICE Directive) on the top toolbar.
- In the dialog box, paste the following command exactly: .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 len=100)
- Place this text anywhere on your schematic area. This defines MyCo-axCable as a 100-meter lossy line.

Step 3: Wiring and Grounding

- Press 'G' to place grounds. You must ground the bottom terminal of the Voltage Source, the two bottom terminals of the ltline (O1), and the Load Resistor.
- Press 'W' to wire the components. Connect all points properly.
- Use 't' to label the input wire as IN and the output wire as OUT.

Step 4: Setting up the Simulation

- Click configure analysis (.ac) from the toolbar or type 'A' to configure transient analysis.

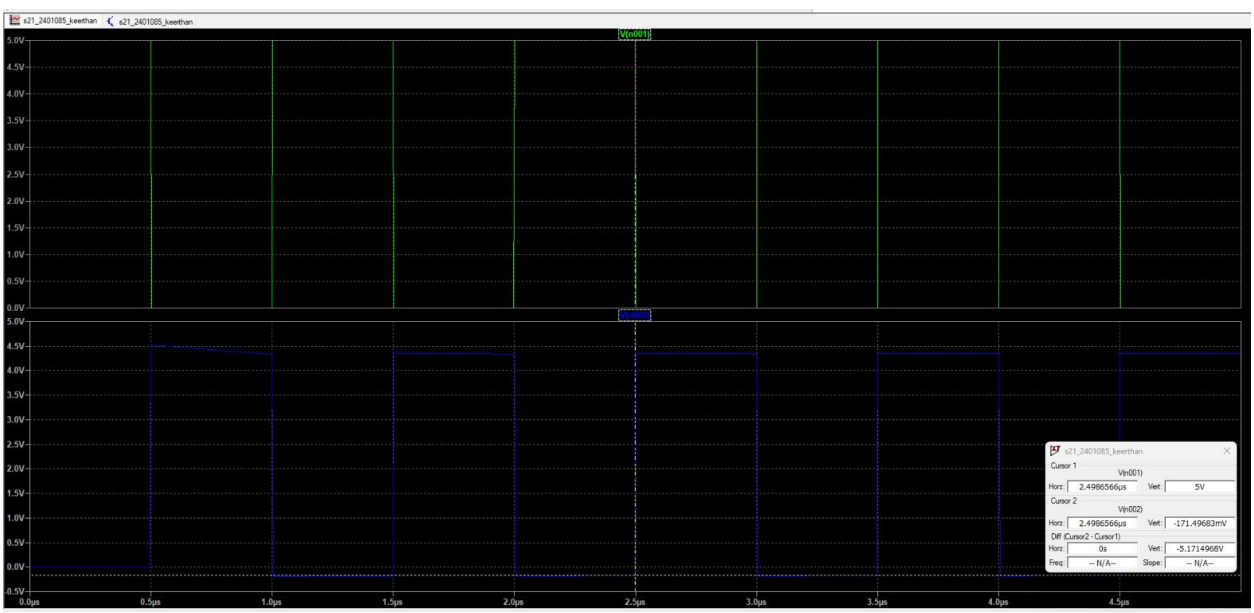
- In the Transient tab, set the Stop Time to 5u.
- Place the .tran 5u command anywhere on the schematic.
- Click the Run (Running Man) icon.

Observation and Analysis

Students must perform the simulation for the following three cases by right-clicking the .model directive and changing the len value each time.

Case 1: Cable Length = 100 meters

- Task: Run simulation with len=100.

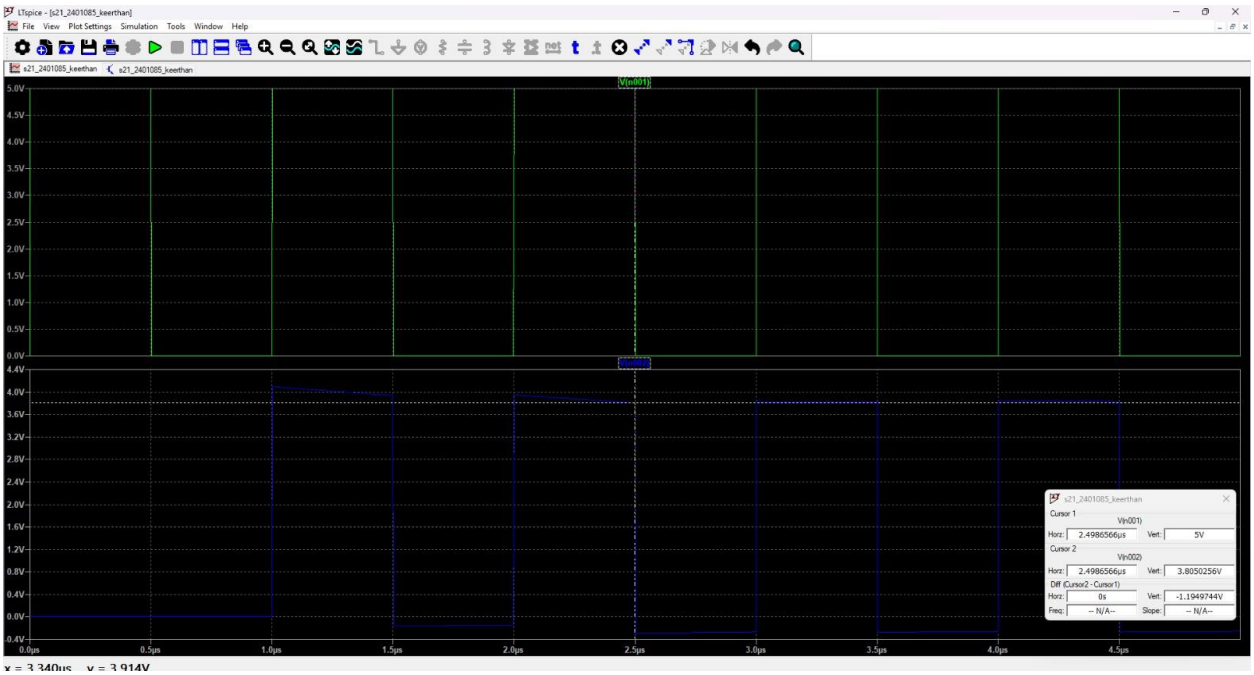


Data: Record the peak output voltage and the time delay.

Case 2: Cable Length = 200 meters

- Task: Update directive to len=200 and re-run. Change the SPICE Directive to define cable length of 200 meters

  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 ***len=200****)*

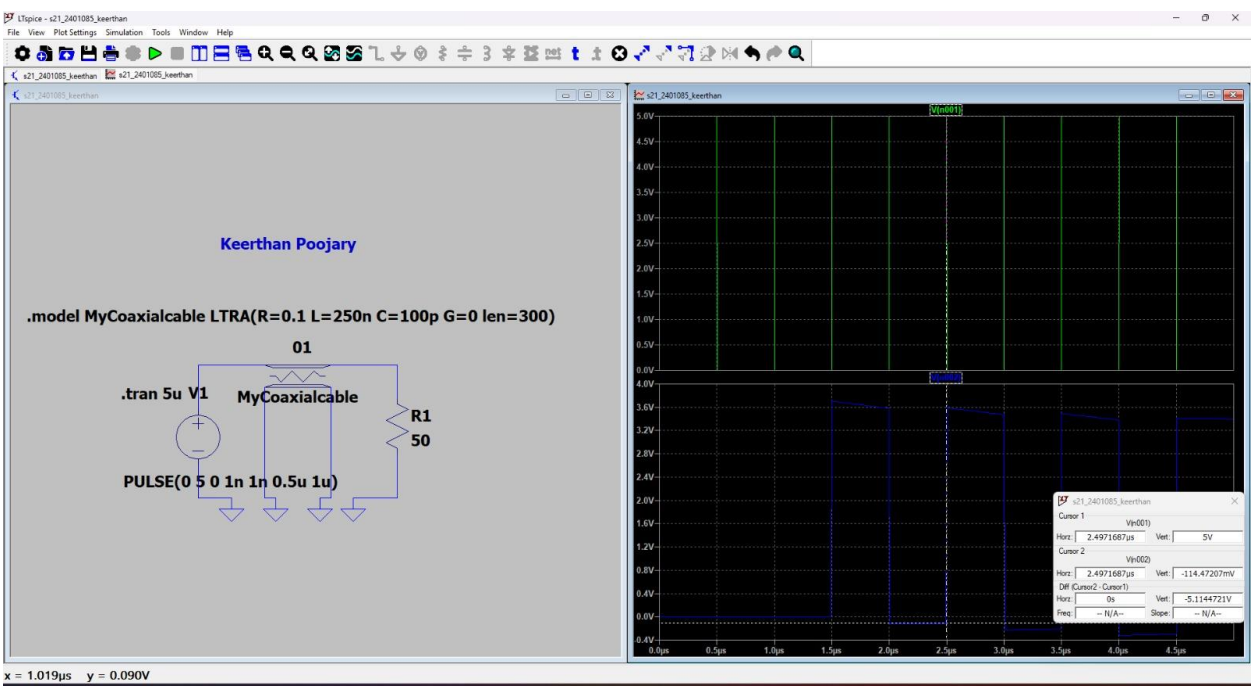


- Data: Record the peak output voltage and observe the increased delay.

Case 3: Cable Length = 300 meters

- Task: Update directive to len=300 and re-run. Change the SPICE Directive to define cable length of 300 meters

  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 ***len=300****)*



- Data: Record the peak output voltage and observe the increased dealy.

Observation Table

Case	Cable Length	Input Voltage	Output Voltage	Propagation Delay
1	**100 m**	**5V**	**2.49 V**	**0.50us**
2	**200 m**	**5V**	**3.804 V**	**1.00 us**
3	**300 m**	**5V**	**1.693 V**	**1.50 us**

Conclusion

The simulation demonstrates that wired media is not ideal. As the cable length increases, attenuation and propagation delay become more prominent. For the cable length of 100 m, 200 m and 300 m the attenuation observed are 59.28%, 23.92%, and 66.14% respectively whereas propagation delay observed was 0.5µs, 1µs and 1.50µs respectively.